

| WORK PACKAGE (WP)                                                                           | PLANNED ACTIVITY (A)                                        | 2027 |     |     |       | 2028  |       |       |       |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|------|-----|-----|-------|-------|-------|-------|-------|
|                                                                                             |                                                             | 1-3  | 4-6 | 7-9 | 10-12 | 13-15 | 16-18 | 19-21 | 22-24 |
| <b>WP1 - Requirements, simulations and design freeze</b>                                    | A1.1 - Requirements definition                              |      |     |     |       |       |       |       |       |
|                                                                                             | A1.2 - SHADOW4 ray-tracing optimization                     |      |     |     |       |       |       |       |       |
|                                                                                             | A1.3 - GEANT4 fluorescence simulations                      |      |     |     |       |       |       |       |       |
|                                                                                             | A1.4 - Final geometry selection                             |      |     |     |       |       |       |       |       |
| <b>WP2 - Room-temperature multichannel CZT spectroscopy for hard X-ray fluorescence XAS</b> | A2.1 - CZT detector and front-end design                    |      |     |     |       |       |       |       |       |
|                                                                                             | A2.2 - Multichannel DAQ and signal-processing development   |      |     |     |       |       |       |       |       |
|                                                                                             | A2.3 - CZT calibration and performance characterization     |      |     |     |       |       |       |       |       |
|                                                                                             | A2.4 - Detector validation and operating-point selection    |      |     |     |       |       |       |       |       |
| <b>WP3 - Instrument integration and commissioning</b>                                       | A3.1 - Procurement of source, optics and motion stages      |      |     |     |       |       |       |       |       |
|                                                                                             | A3.2 - Mechanical design and spectrometer assembly          |      |     |     |       |       |       |       |       |
|                                                                                             | A3.3 - Hardware, software and control-system integration    |      |     |     |       |       |       |       |       |
|                                                                                             | A3.4 - Instrument commissioning and verification            |      |     |     |       |       |       |       |       |
| <b>WP4 - Ag reference samples and realistic sample demonstration</b>                        | A4.1 - Ag foil calibration and reproducibility tests        |      |     |     |       |       |       |       |       |
|                                                                                             | A4.2 - XANES validation on Ag reference compounds           |      |     |     |       |       |       |       |       |
|                                                                                             | A4.3 - Fluorescence XAS demonstration on a realistic sample |      |     |     |       |       |       |       |       |
|                                                                                             | A4.4 - Performance characterization and operating limits    |      |     |     |       |       |       |       |       |